MAIS PAC

IPhO 02.04: Celestial Mechanics and Gravitation · Worked Solutions

Celestial Mechanics and Gravitation: Worked Solutions

MAIS Physics & Astronomy Club (PAC) · Reasoning, derivations and checks

10 worked solutions · independent explanations · consistency checks

Make a genuine attempt before reading. Each solution explains how to choose the method, carries units through the calculation, and checks the result. The exercises are original MAIS PAC teaching problems, not official IPhO questions or marking schemes. Parameters are supplied locally; an exam's constants take precedence.

Solution index

1 2 3 4 5 6 7 8 9 10

Solution 1 — Weighing a planet from its moon. Return →

(a) $T^2 = 4\pi^2 r^3/GM$ $M = 4\pi^2 r^3/(GT^2) = 4\pi^2(1.728\times10^{24})/(6.674\times10^{-11}\cdot1.6\times10^{11}) = 6.4\times10^{23}$ {kg}.

(b) $v = 2\pi r/T = 2\pi(1.2\times10^8)/(4.0\times10^5) = 1.9\times10^3$ {m,s^(-1)}.

(c) $\rho = M/((4)/(3)\pi R^3) = 6.4\times10^{23}/(5.24\times10^{23}) \approx 1.2\times10^3$ {kg,m^(-3)}, comparable to a rocky planet rather than a gas giant. **Check:** a circular-orbit speed of 1.9 {km,s^(-1)} is consistent with a low-mass planet.

Solution 2 — Escape speed and the atmosphere. Return →

(a) $(1)/(2)mv_e^2 = GMm/R$ $v_e = \sqrt{(2GM/R)}$.

(b) Earth: $\sqrt{(2(6.674\times10^{-11})(5.972\times10^{24})/6.371\times10^6)} = 1.12\times10^4$ {m,s^(-1)}. Moon: $\sqrt{(2(6.674\times10^{-11})(7.35\times10^{22})/1.74\times10^6)} = 2.37\times10^3$ {m,s^(-1)}.

(c) $v_{\{rms\}} = \sqrt{(3kT/m)}$: for N_2 ($m=4.65\times10^{-26}$ {kg}) at 300 K this is 518 {m,s^(-1)}, about 1/20 of Earth's escape speed but about 1/5 of the Moon's. With a Maxwellian tail (and the Moon's day-side temperature higher still), a significant fraction exceeds the escape speed over geological time, so the Moon loses gas. **Check:** the ratio $v_e/v_{\{rms\}}$ is the relevant number, not v_e alone.

Solution 3 — Ellipse to circle to escape. Return →

(a) Specific energy before: $\varepsilon_0 = -GM/(2r_0)$. After the burn the speed is αv_c, so $\varepsilon = (1)/(2)\alpha^2(GM/r_0) - GM/r_0 = -(GM)/(2r_0)(2-\alpha^2)$. Comparing with $\varepsilon = -GM/(2a)$ gives $a = r_0/(2-\alpha^2)$.

(b) Escape needs $\varepsilon = 0$, i.e. $2-\alpha^2 = 0$, so $\alpha = \sqrt{2}$.

(c) For $\alpha = 1.2$: $a = r_0/(2-1.44) = 1.786r_0$. Since the burn is tangential and the speed increased, the burn point is periapsis: $r_p = r_0$, so $r_a = 2a - r_p = 2.57r_0$. For $\alpha>\sqrt{2}$ the orbit is hyperbolic; for α slightly less than 1 the burn lowers the periapsis and the orbit dips inside r_0 — a de-orbit burn. **Check:** the $\alpha\to1$ limit returns $a = r_0$, the original circle.

Solution 4 — Kepler's second law, quantitatively. Return →

(a) At periapsis $r_p = a(1-e)$ and $v_p = \sqrt{(GM(1+e)/(a(1-e)))}$, so $h = r_pv_p = \sqrt{(GMa(1-e^2))}$.

(b) Areal velocity is $(1)/(2)r^2\dot{\theta} = (1)/(2)\,h$, constant — Kepler's second law in one line.

(c) From $h = r_pv_p = r_av_a$ with $r_p = a(1-e)$ and $r_a = a(1+e)$: $(v_p)/(v_a) = (r_a)/(r_p) = (1+e)/(1-e)$. **Check:** for $e=0$ the ratio is 1, a circular orbit; for $e\to1$ it diverges, as a near-parabolic orbit should.

Solution 5 — Geostationary orbit and launch cost. Return →

(a) $r = (GMT^2/4\pi^2)^{(1/3)} = (3.986\times10^{(14)}\cdot(86164)^2/39.48)^{(1/3)} = 4.216\times10^7$ {m}.

(b) Height $= 4.216\times10^7 - 6.371\times10^6 = 3.58\times10^7$ {m} $\approx$ 35,800 {km}; speed $2\pi\, r/T = 3.07\times10^3$ {m,s^(-1)}.

(c) Specific energy in GEO: $-GM/(2r) = -4.73\times10^6$ {J,kg^(-1)}; at the surface, $-GM/R = -6.26\times10^7$ {J,kg^(-1)}. The difference, 5.79×10^7 {J,kg^(-1)}, is the ideal launch energy; in LEO the kinetic energy alone is $(1)/(2)\,v_c^2 = 3.1\times10^7$ {J,kg^(-1)}. **Check:** the GEO launch needs about twice the LEO kinetic energy, ignoring losses — orbital velocity dominates every launch budget.

Solution 6 — Tunnelling through the Earth. Return →

(a) Gauss's law for gravity: the mass inside radius r is M(r) = M r³/R³, so g = GM(r)/r² = g_0r/R.

(b) Let x be the distance from the tunnel midpoint. The component of gravity along the tunnel is g_0r/R · (x/r) = g_0x/R toward the midpoint, so x = -(g_0/R)x. Hence ω = √(g_0/R) and T = 2π√(R/g_0).

(c) T = 2π√(6.371×10^6/9.81) = 5060 {s} ≈ 84 {min}. The restoring force per unit mass is -g_0x/R regardless of the tunnel's direction, so the period is the same for every chord. **Check:** a satellite skimming the surface has the same period, which is the deep reason both are 84 minutes.

Solution 7 — Binaries and the barycentre. Return →

(a) r_M = (2M)/(3M)(9.0×10^(10)) = 6.0×10^(10) {m} and r_(2M) = 3.0×10^(10) {m}.

(b) T = 2π√(d³/G(3M)); with M unspecified the period stays in symbolic form, but the ratio r_1/r_2 = 2 is fixed.

(c) v_1/v_2 = r_1/r_2 = 2: the lighter star moves twice as fast. A single-lined binary gives only one velocity amplitude, hence M_2³sin ^3i/(M_1+M_2)² — one equation with two unknowns, so the total mass needs the mass ratio from a second line or from the light curve. **Check:** for equal masses the barycentre is at the midpoint and both speeds are equal.

Solution 8 — The Roche limit. Return →

(a) The tidal acceleration across a body of radius r at distance d is 2GMR_p r/d³, while its self-gravity is G(4πρ_s r³/3)/r² = (4)/(3)π Gρ_s r. Equating them gives d³ (3)/(2π)(M)/(ρ_s), and writing M = (4)/(3)πρ_pR_p³ gives d R_p(2ρ_p/ρ_s)^(1/3).

(b) d 7.15×10^7(2×1330/600)^(1/3) = 7.15×10^7×1.64 = 1.17×10^8 {m}, about 1.6 Jupiter radii.

(c) Ring particles inside that distance cannot accrete into a moon; tidal forces keep shattering any body that tries. **Check:** the limit scales as ρ_s^(-1/3), so a denser body survives closer in — icy comets are torn apart further out than rocky asteroids.

Solution 9 — Gravitational slingshot. Return →

(a) In the planet's frame the encounter is elastic (gravity does no net work over a closed path in that frame, and energy is conserved), so the spacecraft's speed relative to the planet is unchanged; only its direction rotates.

(b) In the planet frame the velocity is u with | u| = u before and after. Transforming back with +U: the final speed along the orbit ranges from |u - U| to u + U depending on the rotation angle, giving the stated bounds with the initial v = |u + U| configuration.

(c) The gain comes from the planet's orbital energy: the planet is slowed by an amount proportional to the spacecraft's mass divided by the planet's mass, which is unmeasurably small but not zero. **Check:** the spacecraft cannot gain energy in the planet's frame, only in the Sun's frame — a distinction that separates a correct answer from a hand-wave.

Solution 10 — Hohmann transfer. Return →

(a) The transfer orbit touches both circular orbits at its apsides, so 2a = r_1 + r_2 and a = (r_1+r_2)/2. At r_1 the required speed is that of the ellipse at periapsis, v_p = √(GM(2/r_1 - 1/a)), compared with the circular speed √(GM/r_1).

(b) Burn 1: Δ v_1 = √(GM(2/r_1 - 1/a)) - √(GM/r_1). Burn 2 at apoapsis: Δ v_2 = √(GM/r_2) - √(GM(2/r_2 - 1/a)). Substituting a = (r_1+r_2)/2 gives the stated expression.

(c) With r_1 = 6.7×10^6 {m} and r_2 = 4.216×10^7 {m}: a = 2.443×10^7 {m}. Burn 1: circular speed 7.71 {km,s^(-1)}, transfer periapsis speed 10.16 {km,s^(-1)}, so Δ v_1 = 2.45 {km,s^(-1)}. Burn 2: transfer apoapsis speed 1.61 {km,s^(-1)}, circular speed at GEO 3.07 {km,s^(-1)}, so Δ v_2 = 1.46 {km,s^(-1)}. Total 3.91 {km,s^(-1)}. Transfer time is half the ellipse period: t = π√(a³/GM) = 1.90×10^4 {s} ≈ 5.3 {h}.

The first burn is larger because the spacecraft must **raise** its energy near the planet, where the potential well is deepest; far away, the same energy change is achieved with a smaller speed change. **Check:** both Δ v are positive for r_2>r_1, and both vanish as r_2→ r_1.